Antenna bias: RC_BIAS, 33 ohm, 47pF to GND, 120R ferrite onto RF_IN. Ferrite is the bias choke, DC is series with the RF path and not on VCC.

J3 X1200 harness: pin 1 to 5: +3V3, GND, SDA, SCL, GPIO6. R7 and R8 I2C pull-ups are DNP.

GPS backup is ML414H directly on V_BCKP, charged from +3V3_GPS through PMEG4005EJ and 1k. CR2032 feeds only RV-3028 VBACKUP.

ICM-45686 0x68, BMP581 0x47, RV-3028-C7 0x52, NEO-M9N UART+PPS, No UPS.

Kingfisher Sensor HAT rev 0.3

ML414H GPS backup, CR2032 RTC, X1200 harness, no UPS ICM-45686 + BMP581 + NEO-M9N + RV-3028	
westphae/kingfisher	
Sheet: /	
File: kingfisher-hat.kicad_sch	
Title: Kingfisher Sensor HAT	
Size: A1 Date: 2026-10-17	**Rev: 0.3**
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